

RIZKI SAPUTRA SEMBIRING

Robotics & Automation Engineer | IoT and Industry 4.0 Specialist

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Passionate engineer with a strong foundation in programming, robotics, and automation. Graduated top of class in Electrical Engineering from Universitas Muria Kudus in 2020. I bring valuable experience from internships at Djarum, where I successfully upgraded a vision sensor system valued at over 1 billion IDR, and Kereta Api Indonesia, where I gained insights into railway infrastructure. Recognized for leading digital transformation and innovation initiatives, I have developed and implemented solutions in computer vision, IoT, automation, and smart factory projects, delivering significant cost reductions and operational efficiencies. I was honored as the Best Employee at Gistex for my contributions to digitalization, innovation, and technology.

EDUCATION

Universitas Muria Kudus

Kudus

Bachelor in Electrical Engineering | GPA: 3.72 / 4.0

2016-2020

Activities: Robotic

- Completed my final project/research to develop cigarette production machine for automatic control at Djarum Oasis Kretek Factory
- Community Service Program in Tondokerto, Pati
- Field work study at LEN Industri & Badan Tenaga Nuklir Nasional

WORK EXPERIENCE

Gistex

Majalengka

Robotic Automation Engineer Supervisor

Mar 2025 – Present

- Lead and supervise teams in the development of technologies including IoT, AMR, robotics, automation and AI computer vision.
- Lead hiring, mentoring, and development of the engineering team, fostering innovation and cross-functional collaboration
- Manage project timelines, budgets, and cross-functional team coordination to deliver solutions on schedule.
- Manage digitization projects in collaboration with cross-functional teams.
- Manage company core project such as automation and robotic arms.
- Collaborate with the R&D team to develop innovative automated sewing machines.

Robotic Automation Engineer Staff

Jan 2022 – Mar 2025

- Develop and deploy computer vision solutions to support company objectives.
- Optimize resource utilization, including hardware and budgets.
- Designed and implemented IoT and automation systems, covering hardware selection, firmware development, and cloud integration.
- Led smart factory and productivity improvement projects, utilizing IoT technologies to enhance operator efficiency and monitor performance.
- Reduced automation and IoT implementation costs by approximately 1 billion IDR compared to market alternatives.
- Built and maintained web and desktop applications, improving UX and functionality.
- Created and optimized databases for enhanced query performance and data integrity.

Kereta Api Indonesia

Bandung

Signalling, Telecommunication, and Electricity Engineer Intern

Mar 2020 – Aug 2020

- Participated in an internship program facilitated by Forum Human Capital Indonesia BUMN.
- Collaborated with the Sintelis core engineering team at Kereta Api Indonesia HQ.
- Gained hands-on experience in electrical-mechanical signalling systems and CBTC (Communication-Based Train Control) technology.
- Learned about the design and operation of railway communication and electrical systems.

Djarum*Research Intern*

Kudus

Jan 2020 – Feb 2020

- Research to develop one of the cigarette tip paper processing production machines into auto system as thesis project

Djarum*Electrical Engineer Intern*

Kudus

Jul 2018 – Sep 2018

- Collaborated with senior engineers to design, install, and maintain control systems for production machinery and processes.
- Conducted testing and tuning of electrical systems, including control panels, sensors, and automation components.
- Successfully upgraded a control panel and vision sensor system to detect defective cigarette excise bands with costs over 2 billion IDR.
- Gained practical experience in electronic control systems, instrumentation, and automation within one of Indonesia's largest tobacco industries.

NON-FORMAL EDUCATION

Cisco Networking Academy*CISCO*

Kudus

2013 – 2015

- Focused on designing, configuring, and troubleshooting network infrastructures for small-to-medium businesses and ISPs.
- Developed expertise in IP addressing, VLANs, routing protocols, and network security for scalable solutions.
- Covered foundational networking concepts, including designing, and implementing secure wired and wireless networks for homes and small businesses.
- Emphasized practical skills in troubleshooting, network optimization, and securing data transmissions.

CERTIFICATION

- FreeRTOS, PT. Nusabot Inovasi Teknologi (2024)
- Program Magang Mahasiswa Bersertifikat, Forum Human Capital Indonesia (2020)
- CCNA Discovery: Working at a Small-to-Medium Business or ISP, CISCO (2015)
- CCNA Discovery: Networking for Home and Small Businesses, CISCO (2014)

AWARDS

- Best Employee (2024): Recognized for contributions to digitalization, innovation, and technology at Gistex
- Best Graduate (2020): Graduated as the top student in Electrical Engineering
- Academic Achievement Scholarship (2018): 1-year academic development scholarship funded by the Ministry of Research, Technology, and Higher Education for the best parallel students
- Highest Test Score of Electrical Engineering in the admission of new students (2016)

SKILL

Soft Skill

- Innovation, Problem-Solving, Leadership, Teamwork, Project Management, Communication.

Technical Skill

- OpenCV, Robotic, Internet of Things, Automation, Embedded Programming, Database Management, Instrumentation and Control System, Smart Factory and Digital Transformation, Microcontroller, MQTT

Software Tools

- Arduino, Visual Studio, GitHub, MySQL, C++, C# .Net, Python, PHP, HTML, Bootstrap

LANGUAGES

- English (Professional)
- Indonesian (Native)